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A comparison between Multilayer Perceptrons and Kolmogorov-Arnold Networks for multi-task classification in sitting posture recognition

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ABSTRACT Prolonged incorrect sitting postures can lead to various health issues, including musculoskeletal disorders and reduced quality of life. Efficient and accurate methods for monitoring and assessing sitting postures are crucial for promoting overall well-being in our increasingly sedentary society. This study compares the performance of Kolmogorov-Arnold Networks (KANs) and Multilayer Perceptrons (MLPs) in sitting posture recognition, utilizing a multi-task classification approach to simultaneously identify upper and lower body positions. A comprehensive dataset of sitting postures was developed using MediaPipe for skeletal extraction, labeled for both body regions, and encompassing a range of common scenarios. The models were implemented and evaluated using Leave-One-Subject-Out (LOSO) cross-validation to assess generalization capability across different individuals. Results indicated that the KAN model achieved higher accuracy (97.03% for upper body and 92.11% for lower body) with fewer parameters (4,320) compared to the MLP (93.87% and 91.47% respectively, with 7,662 parameters). Although the MLP demonstrated faster inference, both models achieved accuracies exceeding 91% in LOSO validation, suggesting their potential for real-time applications. This study significantly contributes to the field of automatic posture recognition, offering insights into the applicability of KANs and MediaPipe-based pose estimation in ergonomic recognition systems and demonstrating the effectiveness of a multi-task approach for comprehensive sitting posture analysis.

INDEX TERMS Ergonomics, Kolmogorov-Arnold Networks, MediaPipe, Multi-task Classification, Posture Recognition

I. INTRODUCTION

IN our increasingly sedentary society, maintaining proper posture is crucial for preventing musculoskeletal disorders and promoting overall well-being [1], [2]. Prolonged periods in incorrect postures, whether at a desk or on a couch, can lead to numerous health issues, including back pain, neck strain, and reduced quality of life [3]–[6]. Consequently, there is a growing need for efficient and accurate methods to monitor and assess sitting postures in both professional and personal settings.

Ergonomics, traditionally focused on workplace optimization [7], has expanded to address postural concerns in various aspects of daily life [8]. Identifying and correcting incorrect postures can significantly reduce health risks and improve comfort [9]. However, conventional ergonomic assessments often rely on manual observations or expensive specialized

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equipment, limiting their widespread application [10].

Recent advancements in computer vision and machine learning have opened new possibilities for automated posture recognition systems. These systems offer the potential for continuous, non-invasive monitoring of sitting postures in various contexts, from office spaces to living rooms [11]. Among the available tools for skeleton extraction from images, MediaPipe has gained popularity due to its efficiency and cross-platform compatibility, making it suitable for a wide range of applications [12], [13].

While Multi-Layer Perceptrons (MLPs) and Convolutional Neural Networks (CNNs) have been widely used for posture classification tasks [14], [15], the recent introduction of Kolmogorov-Arnold Networks (KANs) presents a promising alternative [16]. KANs replace traditional weight matrices with trainable b-splines, potentially offering advantages in parameter reduction, generalization, and interpretability. However, the applicability of KANs in real-world posture recognition scenarios, especially in diverse everyday settings and for multi-task classification, remains largely unexplored.

This paper aims to advance the state-of-the-art in posture recognition through several innovations. By comparing MLPs and KANs, we seek to understand whether the theoretical advantages of KANs translate into practical benefits in real-world applications. Our approach uniquely employs a single network to simultaneously classify both upper and lower body postures, providing an ideal scenario to evaluate how both architectures handle multiple related classification tasks. Furthermore, we implement a Leave-One-Subject-Out cross-validation methodology, providing a more rigorous evaluation of model generalization than conventional approaches. The main contributions of this study are:

1) Development of a comprehensive dataset of sitting postures, labeled for both upper and lower body positions, encompassing a range of common scenarios.
2) Implementation and comparison of MLP and KAN models for multi-task posture classification, specifically addressing the simultaneous recognition of upper and lower body postures.
3) Evaluation using Leave-One-Subject-Out (LOSO) cross-validation to assess generalization capabilities across different individuals.
4) Analysis of model performance in terms of accuracy, parameter efficiency, and inference speed, considering the practical requirements of real-world applications and the challenges of multi-task classification.

By focusing on these aspects, we seek to provide valuable insights into the practical applicability of KANs in ergonomic posture recognition systems. Our goal is to pave the way for more efficient and accurate posture monitoring solutions, ultimately contributing to improved health and well-being in our increasingly sedentary lifestyles.

The remainder of this paper is structured as follows: Section II provides a review of related work in sitting posture recognition across various settings. Section III outlines the methodology, encompassing dataset creation, model design, and evaluation procedures. Section IV presents experimental results. Section V offers a discussion on the implications of the findings and a comparison with existing literature. Finally, Section VI concludes the study and proposes directions for future research in diverse applications of posture recognition.

II. RELATED WORK

Recent advancements in computer vision and machine learning have led to significant improvements in sitting posture recognition systems. This section reviews key studies in the field, highlighting their approaches, contributions, and limitations in addressing the challenges of developing efficient and accurate posture monitoring solutions.

Camera-based systems have been a popular choice due to their non-invasive nature and ability to capture detailed posture information. Chen [17] developed a sitting posture recognition system for classroom environments using OpenPose and a convolutional neural network (CNN), achieving over 90% accuracy after 100 training epochs. However, this study only distinguished between two postures (correct and incorrect) and relied solely on training accuracy for evaluation, potentially limiting its real-world applicability. Similarly, Estrada et al. [18] proposed a system for classifying proper and improper sitting postures in work-from-home settings using smartphone and webcam cameras. Their approach, combining MediaPipe for pose estimation with decision trees for classification, achieved accuracies of 85.18% and 92.07% on two datasets. While these camera-based systems show promise, they raise privacy concerns and often lack the granularity needed for comprehensive posture analysis.

To address some of these limitations, researchers have explored more advanced neural network architectures and alternative sensing technologies. Jiao et al. [19] introduced LMSPNet, a lightweight network for multi-person sitting posture recognition, achieving an impressive 99.57% accuracy on a custom dataset of three postures. Although this study addresses computational efficiency, the limited number of postures and lack of cross-subject validation raise questions about its generalizability. In contrast, Gupta et al. [20] designed an autonomous sitting posture recognition system using 15 pressure-sensitive conductive (PSC) sensors on an ergonomic chair. Their CNN model outperformed traditional machine learning approaches in classifying eight different postures, demonstrating the potential of sensor-based approaches in capturing more nuanced posture information. Similarly, Tsai et al. [21] developed a pressure sensor-based system that achieved 99.1% accuracy in recognizing ten different sitting postures, while also providing comprehensive validation through user satisfaction metrics, although their approach still shares the common limitation of requiring specialized hardware.

Privacy concerns have led to the exploration of alternative sensing technologies. Lin et al. [22] developed a chair system using two LiDAR cameras to capture depth images for

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posture detection, achieving 99.71% accuracy with an EfficientNet Lite2 model. While this approach preserves privacy, the use of LiDAR technology may be prohibitively expensive for widespread adoption. Taking a different approach, Xu and Liu [23] proposed using Frequency Modulated Continuous Wave (FMCW) radar for posture detection in bathroom environments, achieving 98.75% accuracy. This method effectively preserves privacy but may not capture the fine-grained details necessary for comprehensive posture analysis.

The diversity of approaches in the field was highlighted in a comprehensive review by Odesola et al. [24], which noted that while force-sensitive resistors (FSRs) are commonly used in smart sensing chairs, vision-based systems with CNNs and ANNs are emerging as powerful tools. However, they emphasized that these advanced models often do not outperform traditional statistical models due to limited dataset sizes and lack of diversity, underscoring the need for more robust data collection and validation methodologies.

Recent innovations have also explored novel data inputs and model architectures. Nishitha et al. [25] compared CNN and Region-based CNN (RCNN) for sitting posture analysis, achieving accuracies of 92.5% and 95% respectively. While demonstrating the potential of region-based approaches, the study's lack of clear validation methodology and limited dataset information raises questions about the robustness of these results.

While existing studies have made significant progress, they often rely on large networks, classify a limited number of postures, or use expensive sensors like LiDAR or pressure-sensitive arrays. Many lack a comprehensive approach to upper and lower body posture recognition, and suffer from limited interpretability and poor generalization across individuals. There remains a need for efficient, multi-task models capable of classifying a wide range of postures using affordable sensors like standard cameras, while offering insights into their decision-making process. Addressing these gaps could lead to more practical, cost-effective, and widely applicable posture recognition systems for real-world settings.

III. MATERIALS AND METHODS

This section outlines the methodological approach used in our study, encompassing dataset creation, model design and selection, and evaluation procedures.

A. DATASET CREATION

The development of a comprehensive dataset was essential to effectively train and evaluate the proposed sitting posture recognition system. For the creation process, we have followed the three phases explained below: collection, preprocessing and manual labelling.

1) Data collection

To create this dataset, 13 volunteers participated in the data collection process. The participants had a mean age of 32.46 ± 13.39 years, representing a diverse adult population. All participants signed an informed consent form prior to their involvement in the study. This consent form detailed the nature of the data collection process and the intended use of the data. The data collection setup involved positioning a camera in front of each participant while they were seated on a chair or sofa. This setup aimed to simulate typical everyday environments where posture recognition would be applicable.

Participants were instructed on how to perform each posture, but were encouraged to execute them in a way that was natural and representative of their everyday behavior. This approach was chosen to improve the validity of the data set and to ensure that the trained models were effective in real situations. The resulting dataset is comprehensive and complex, comprising a total of 4,800 frames. Each participant performed multiple iterations of various sitting postures, contributing to a rich and diverse collection of data that captures a wide range of natural sitting behaviors and variations.

The experimental setup employed a Reolink 5MP RLC-510A[1] camera positioned facing the participants at a distance of 1 to 2 meters. The camera's position allowed for slight lateral adjustments while capturing full-body seated postures. This configuration ensured clear recording of various sitting positions while maintaining a naturalistic environment. To protect participant privacy, the raw images captured during this process cannot be distributed. Only the processed skeletal data extracted from these images is made available for research purposes, ensuring participant anonymity while still providing valuable data for posture recognition studies.

The positions performed are as follows:

1) Upright posture
2) Leaning backward
3) Leaning forward
4) Right lateral tilt
5) Left lateral tilt
6) Right leg crossed over left
7) Left leg crossed over right
8) Legs spread apart
9) Legs closed

Figure 1 provides visual examples of these postures as performed by the participants. This figure illustrates four of the nine sitting postures examined in the study: (a) Left leg crossed over right, (b) Leaning forward, (c) Right lateral tilt, and (d) Leaning backward.

This diverse set of postures was selected to encompass a wide range of common sitting positions observed in everyday settings. By including both symmetric and asymmetric postures, as well as variations in upper and lower body positioning, the dataset aims to capture the complexity of real-world sitting behaviors [26], [27].

2) Data preprocessing

The data preprocessing phase began with the extraction of individual frames from the recorded videos. These videos,

[1] https://reolink.com/es/product/rlc-510a/

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(a) Left leg crossed over right. (b) Leaning forward. (c) Right lateral tilt. (d) Leaning backward.

FIGURE 1: Examples of sitting postures performed by study participants.

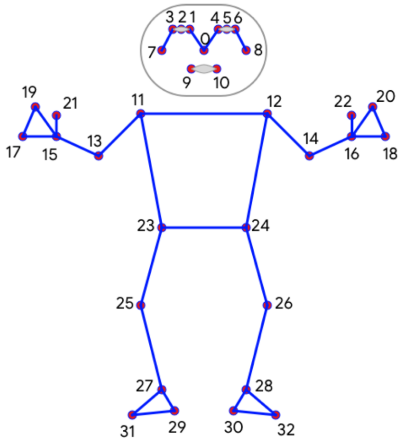


FIGURE 2: Keypoint topology used by BlazePose [28].

originally captured at 2K resolution, contained various desired postures. To optimize computational efficiency while retaining sufficient detail for pose estimation, the extracted frames were subsequently downscaled to a resolution of 640x480 pixels. This scaling process struck a balance between image quality and processing requirements, ensuring that the subsequent pose estimation could be performed effectively.

For skeleton extraction, the MediaPipe Pose Heavy model was utilized. MediaPipe Pose is based on the BlazePose architecture [28]. MediaPipe was selected over alternatives like OpenPose due to its efficiency, cross-platform compatibility, and ability to provide 3D pose estimation. The Heavy model variant was chosen for its higher accuracy, which is crucial for detailed posture analysis. Figure 2 illustrates the topology of keypoints used in the BlazePose model.

From the MediaPipe output, the World Landmarks were specifically extracted. These provide three-dimensional coordinates in real-world meters, with the origin set at the midpoint of the hips. This choice of coordinate system allows for consistent pose representation regardless of camera position or subject size.

Figure 3 illustrates the result of applying MediaPipe Pose to the frames shown in Figure 1, demonstrating the skeletal representation extracted from the original images.

To prepare the data for model input, a StandardScaler was applied to normalize the feature values. Furthermore, to reduce the dimensionality of the input from 99 (33 joints x 3 coordinates) to 33 (11 joints x 3 coordinates), the arm joints and all facial landmarks except for the nose point were eliminated. This dimensionality reduction focuses the model on the most relevant features for posture recognition while reducing computational complexity.

3) Data labelling

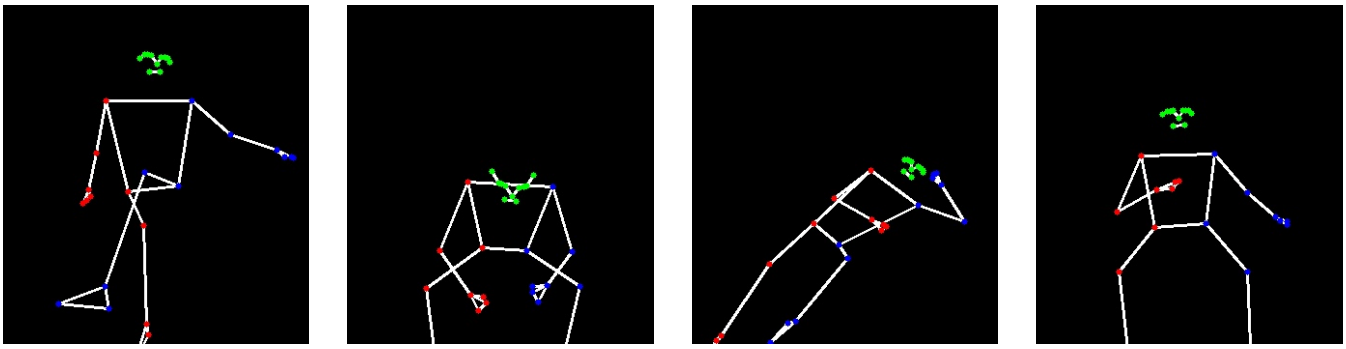
The data labeling process was designed with consideration for established ergonomic principles for working posture. According to the international standard ISO 11226-2000 for the evaluation of ergonomic aspects of working postures, four main body parts compose the sitting position: the trunk, the head, the upper extremity, and the lower extremity [29]. This standard considers an upright sitting position as optimal because it distributes force evenly over both hips and minimizes lumbar compensation [30]. While the ISO standard divides the body into more segments than we use in our study, we focused primarily on the trunk and legs, as we are not evaluating a work environment where upper extremity positioning would be crucial.

Based on these considerations and the specific context of our study, it was decided to classify postures into two main categories: upper body posture and lower body posture. This division allows for a more detailed and specific analysis of the key components of the sitting posture, focusing on the body parts that have the greatest impact on ergonomics and overall well-being.

Consequently, each instance in the dataset has two associated labels: one for lower body posture and one for upper body posture. This dual labeling approach facilitates a more granular analysis of postures and allows machine learning models to capture the nuances of different combinations of trunk and leg postures. The labeling process was conducted

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(a) Left leg crossed over right. (b) Leaning forward. (c) Right lateral tilt. (d) Leaning backward.

FIGURE 3: Examples of sitting postures performed by study participants.

manually, with each frame undergoing multiple rounds of review to ensure accuracy and consistency. The specific labels used for each category are shown in Table 1 and 2.

TABLE 1: Upper body posture labels.

Upper Body Posture Code	Description
TUP	Upright
TLB	Lean Backward
TLF	Lean Forward
TLR	Lean Right
TLL	Lean Left

TABLE 2: Lower body posture labels.

Lower Body Posture Code	Description
LAP	Apart
LWA	Wide Apart
LCS	Closed
LCR	Crossed Right
LCL	Crossed Left
LLR	Lateral Right
LLL	Lateral Left

The resulting labeled dataset, which includes the skeletal keypoint coordinates along with their corresponding posture labels, is available in the supplementary material accompanying this article. This resource provides a valuable foundation for future studies in the field of automatic posture recognition in domestic environments.

B. MODEL DESIGN AND SELECTION

ANNs are machine learning models that can be trained using data labelled to obtain a system capable of predicting the label of new unlabelled data[2]. These models can have very complex architectures with many layers and connections, so designing them is a non trivial process, and often it requires finding the best hyperparameters using automated or manual techniques. For this work, two different neural networks (a MLP and a KAN) have been defined, trying to obtain the best results of each of them for the same evaluation method.

1) Multilayer perceptron

A MLP [31] is a type of feedforward neural network architecture composed by a set of linear layers followed by a non-linear function. Each of those linear layers is form by neurons, that represent a weighted sum of the output of the neurons from the previous layers. Each of these weights can be trained using the backpropagation technique [32] which uses gradient descent approaches to minimize the outcomes of a loss function computed using the expected result and final output of the network. The final goal of the training is obtaining a function that, given an unlabelled data input of the same type of the one used to train the model, can produce a new label.

For classification problems in MLPs one hot encoding is usually used. This consists of transforming a single variable that can contain up to C values, each one representing a single class, into an array of C variables where only one has a value of 1 and the rest have a value of 0. The position of the variable with value 1 corresponds with the index of the original label. This transformation is fundamental to being able to train these kind of models, while making simple comparing the result with the output of the network, where each neuron of the last layer is related with a position of the encoded label.

As detailed in Section III-A3, this work employs a multi-task learning approach to predict labels for both lower and upper parts of the body simultaneously. In our approach, we have decided using a single model to predict two labels, which is called in the literature as multi-task labelling [33]. What characterizes multi-task learning is that, despite using the same network, each task loss is computed differently. However, considering how backpropagation works, obtaining a single loss value is more convenient, so losses function must be combined some way. The loss from each task i is computed using Cross Entropy as depicted in Equation 1, and Equation 2 shows how final loss function is computed.

[2]ANNs can also be used for regression and other types of tasks, but in this paper they are only used for class labelling purposes, and, therefore, the text is focused only on that.

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$$l(y, y', C) = -\sum_{c=0}^{C-1} \ln \frac{e^{y'[c]}}{\sum_{i=0}^{C-1} e^{y'[i]}} \cdot y[c] \tag{1}$$

$$L = l(y_l, y'_l, C_l) + l(y_u, y'_u, C_u) \tag{2}$$

, where y_l and y_u are arrays containing the one-hot encoded label of lower and upper parts of the body respectively, y'_l and y'_u are the output of the last neural network layer for each body set of parts, and C_l and C_u are the number of classes of each label. Summation of losses is a simple but effective way of combining them, but it may have the problem of prioritizing one loss over the other. In this case, this problem was not observed, so the summation of single task losses have been kept as final loss.

Figure 4a shows the architecture of the MLP based network used in this work. That configuration have been obtained after performing the hyperparameter tuning procedure described in III-B5. The MLP model have been implemented it using Torch [34].

2) Kolmogorov-Arnold Network

KAN architecture is inspired by MLPs and the Kolmogorov-Arnold representation theorem [35], which states that any multivariate function on a bounded domain can be written as a composition of continuous univariate functions and summations. This theorem inspired the paper of Liu et al. [16] in which they propose a new architecture for ANNs. In summary, this architecture replaces weights and fixed nonlinear functions in MLPs with trainable activation functions that have as core b-splines.

A layer in a KAN is defined by two parameters: the input dimension (n_{in}) and the output dimension (n_{out}). It is conformed by $n_{\text{in}} \times n_{\text{in}}$ activation functions, that can be arranged as a matrix Φ as Equation 3 shows.

$$\Phi = \phi[p, q]; \quad p \in [1, n_{\text{in}}]; \quad q \in [1, n_{\text{out}}] \tag{3}$$

The forward step is defined in Equation 4.

$$\text{forward}(\mathbf{x}) = \begin{pmatrix} \phi_{[1,1]}(\mathbf{x}_{[1]}) & \phi_{[1,2]}(\mathbf{x}_{[2]}) & \dots & \phi_{[1,n_{\text{in}}]}(\mathbf{x}_{[n_{\text{in}}]}) \\ \phi_{[2,1]}(\mathbf{x}_{[1]}) & \phi_{[2,1]}(\mathbf{x}_{[2]}) & \dots & \phi_{[2,n_{\text{in}}]}(\mathbf{x}_{[n_{\text{in}}]}) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{[n_{\text{out}},1]}(\mathbf{x}_{[1]}) & \phi_{[n_{\text{out}},2]}(\mathbf{x}_{[2]}) & \dots & \phi_{[n_{\text{out}},n_{\text{in}}]}(\mathbf{x}_{[n_{\text{in}}]}) \end{pmatrix} \cdot O_{n_{\text{in}}} \tag{4}$$

, where $O_{n_{\text{in}}}$ is a vector that only contains ones of size n_{in}, that is used to add together each value of each row.

Each activation function ϕ is defined by Equation 5.

$$\phi(x) = w \cdot (\text{SiLU}(x) + \text{spline}(x)) \tag{5}$$

, where SiLU is the Sigmoid Linear Unit function [36], and spline is a b-spline. A b-spline is a curve created by the concatenation segments built as polynomial functions of order n defined by a set of control points, also known as Boor points [37]. These control points are the trainable parameters of a KAN, and how many of them are is defined by two hyperparameters: grid size and spline order. The other main hyperparameters that define the architecture of a KAN would be the number of layers and the number of neurons of each layer.

The architecture proposed for our KAN model is also designed to perform multi-task classification, and, therefore, uses the same loss function than the MLP described above, and represents the labels using one-hot encoding. Figure 4b shows the architecture of the KAN used in this work, that as with MLPs, has been obtaining after performing the hyperparameter tuning procedure described in III-B5. The implementation of the model has been done using efficient-kan library [38], which modifies the original library provided by the authors of the KAN paper [16] to increase its speed.

3) Training Process

For both MLP and KAN models, a similar training process was employed to ensure a fair comparison. The training was conducted on a system equipped with an NVIDIA RTX 3050 GPU, 16 GB of RAM, and an AMD Ryzen 5 5600H processor with 6 cores.

The AdamW optimizer was utilized for both models. This optimizer is an extension of the Adam optimizer that incorporates decoupled weight decay regularization, which can help in reducing overfitting [39].

To further optimize the learning process, the ExponentialLR scheduler was implemented. This scheduler exponentially reduces the learning rate over time, allowing for larger updates at the beginning of training and smaller updates as the model converges.

The training process for both models consisted of 10 epochs.

The specific hyperparameters for each model, including learning rate, weight decay, and gamma (for the ExponentialLR scheduler), were determined through the hyperparameter tuning process described in section III-B5.

During training, both the training loss and validation metrics were monitored to ensure the models were learning effectively and not overfitting. The LOSO cross-validation strategy, as detailed in section III-B4, was employed to validate the models' performance and generalization capability.

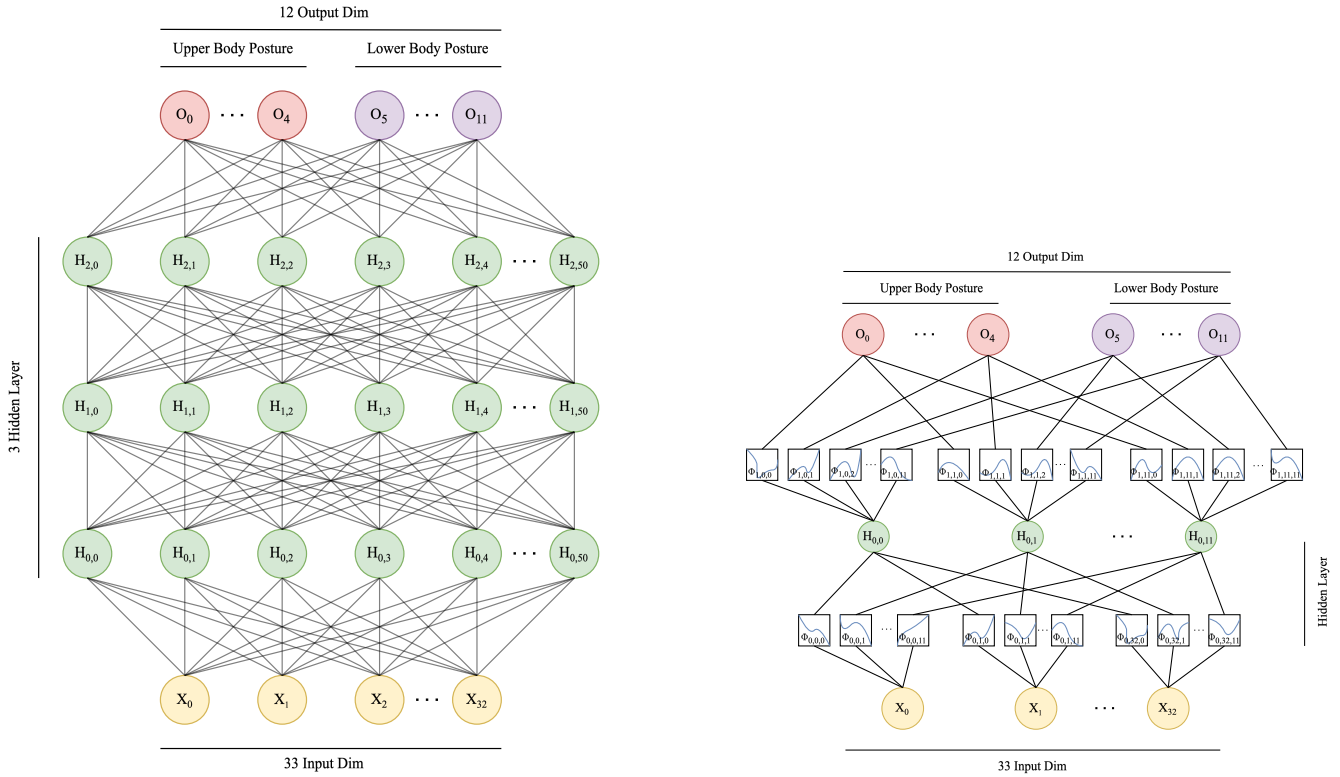
4) Model evaluation

A proper evaluation of the performance of a model is fundamental to measure how good it is. In this work, two aspects of the model are evaluated: classification quality and inference speed.

Cross-validation technique is one of the most powerful tool to estimate classification quality. It consists of creating multiple train/test splits on the original data set, training the model using the train splits, evaluating it with the test splits by computing different metrics, and averaging those metrics for each split. However, using the correct split is fundamental to avoid bias in the validation results. A typical technique for splitting is k-fold cross-validation, that separates randomly the dataset in k subsets, and then, for each of those subsets one is selected as test set, and the rest are combined into

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(a) Multilayer perceptron architecture.

(b) Kolmogorov-Arnold network architecture.

FIGURE 4: Comparison of model architectures: (a) Multilayer perceptron (MLP) with three hidden layers, and (b) Kolmogorov-Arnold network (KAN) with a single hidden layer. Both architectures have 33 input dimensions corresponding to skeletal keypoints and 12 output dimensions representing upper and lower body posture classifications.

a training set. However, traditional k-fold cross-validation is not suitable for our dataset structure. Since we collected multiple samples from each participant performing similar postures, random splitting could result in having highly correlated samples from the same person in both training and test sets, leading to overly optimistic performance estimates.

To ensure a more rigorous evaluation of the model's ability to generalize to new users, we implemented LOSO cross-validation. In this approach, instead of random splits, we iteratively use data from one participant as the test set while training on data from all other participants. This process is repeated for each of our 13 participants, effectively testing how well the model performs on completely unseen individuals. The metric computed for the evaluation is accuracy, defined in Equation 6, which works fine for multi-class problems.

$$\text{accuracy} = \frac{C}{T} \tag{6}$$

, where C is the number of correct predictions and T is the total number of predictions.

The classification speed is been also measured during the cross-validation phase, obtaining an average frame inference time for each participant. Later, the average time for each participant has been averaged again to obtain the final mean classification time. Times are measured on CPU, unlike training that used GPU. To ensure consistent measurements, the same computer used for training have been used for both models speed measurements.

5) Hyperparameter Tuning

The term hyperparameter refers to a parameter that is set for a model that condition the training process. Most typical hyperparamers for ANNs are either architecture related, like the number and size of layers, or learning related, such as learning rate or weight decay. These hyperparameters affect models performance, both in time and classification quality. Therefore, selecting the proper hyperparameters is crucial for any machine learning task. There are many methods for hyperparameter tuning, being the simpler manually modifying them and then checking how that affects the results. Other typical technique is grid search, that tests using brute force all the combinations between different sets of predefined values for each hyperparameter, which is a really slow process.

Luckily, there are other more efficient and useful techniques suitable for our needs, such as the Non-Dominated Sorting Genetic Algorithm 2 (NSGA-II) [40]. This algorithm can be used to find solutions for a multi-objective optimization problem. In this paper, Optuna library [41], which

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implements this algorithm, has been used for hyperparameter tuning. The tool uses the Pareto front to provide a list of the best combinations of hyperparameters. To choose the definitive combination, authors manually select one from that list, trying to maximize the accuracy while keeping a low number of parameters.

We have established three optimization objectives for our models: LOSO upper body accuracy, LOSO lower body accuracy, and model size in number of parameters. Hyperparameters selected for each model to be optimized are shown in Table 3.

TABLE 3: List of hyperparameters used in KAN and MLP models.

Common Hyperparameters	
Learning rate. The step size that determines how much model parameters are updated during optimization.	
Weight decay. A regularization technique that penalizes large weights to prevent overfitting.	
Gamma. The decay factor in the ExponentialLR scheduler that exponentially reduces the learning rate.	
MLP	**KAN**
Hidden layers size. The number of neurons for every hidden layer.	**Hidden layers size**. The number of neurons in the hidden layer.
Number of hidden layers. The quantity of hidden layers in the model.	**Grid size**. The number of segments in the spline.
	Spline order. The degree of curves used in each segment of the spline.

IV. RESULTS

The experimental results are presented in three main parts: hyperparameter optimization, model accuracy evaluation, and computational efficiency analysis. This section begins with the outcomes of the hyperparameter tuning process for both the MLP and KAN models.

Figure 5 contains two graphs, one for the KAN model and another for MLP model, that depict the result of the hyperparameter tuning process by plotting scatter diagram highlighting the Pareto front and the selected combination of parameters. Given that there are three objectives to optimize, the graph should have three axis, but, for the sake of better visualization within the article, lower body and upper body accuracy values have been merged by averaging them into a single accuracy. Table 4, shows the final selected values for each hyperparameter and each model, which correspond to the red points in Figure 5. They have been chosen manually from the Pareto front of the results trying to achieve high accuracy values while keeping small models.

The hyperparameters chosen (represented by the red points in Figure 5 and detailed in Table 4) yielded the following results in the process of maximizing lower body accuracy, maximizing upper body accuracy, and minimizing the number of parameters. For the KAN model, the lower body accuracy was 96.29%, the upper body accuracy was 93.12%, and the

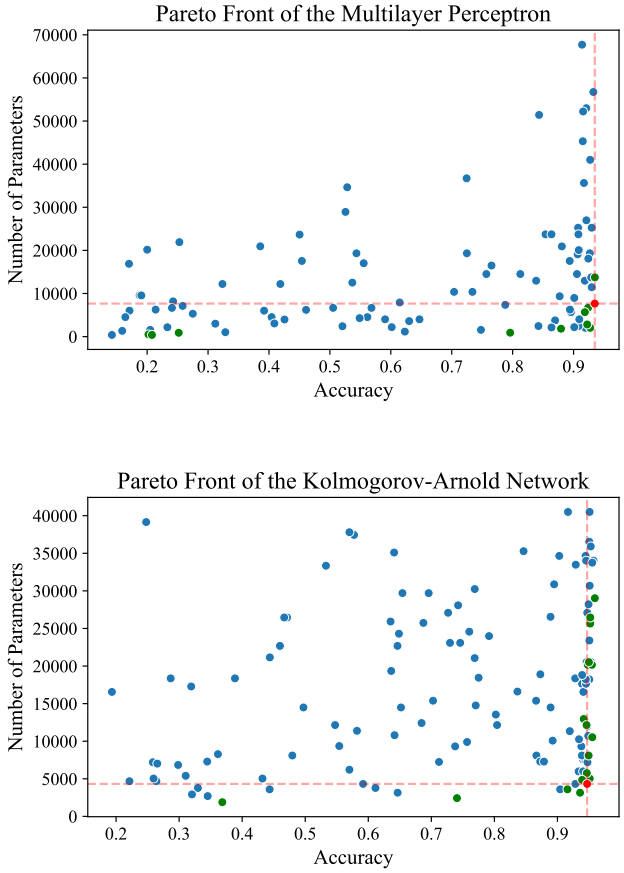


FIGURE 5: 2D representation of a 3D Pareto front that shows the mean accuracy between upper body and lower body predictions against number of parameters. Green points are inside the front, and red points correspond to the chosen parameters.

TABLE 4: Results of hyperparameter search for KAN and MLP models.

Hyperparameters MLP	**Value MLP**	**Hyperparameters KAN**	**Value KAN**
Learning rate (lr)	0.001	Learning rate (lr)	0.006
Weight decay	1.759e-04	Weight decay	3.321e-6
Gamma	0.984	Gamma	0.856
Hidden dim	51	Hidden dim	12
Number of layers	3	Grid size	3
		Spline order	3

number of parameters was 4320. For the MLP model, the lower body accuracy was 95.01%, the upper body accuracy was 91.96%, and the number of parameters was 7662

Figure 6 presents the accuracy results for both the MLP and KAN models using the LOSO cross-validation method. The graphs display the prediction accuracy for upper and lower body postures for each of the 13 participants when their data is used as the test set. For the MLP model, upper body accuracy ranges from 85.56% to 100%, with a mean of $93.87\%(\pm 4.17\%)$, while lower body accuracy spans from 72.22% to 99.67%, averaging $91.47\%(\pm 8.58\%)$.

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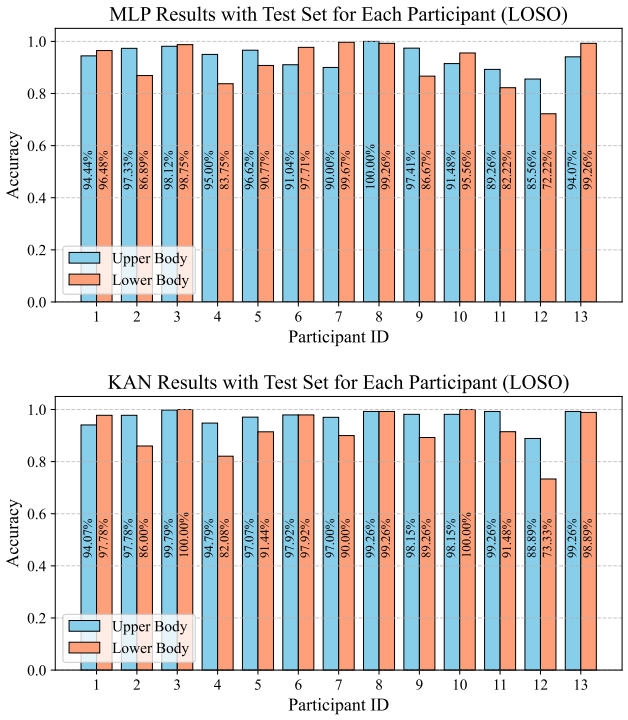


FIGURE 6: Comparison of MLP and KAN model performance using LOSO cross-validation. The graphs show the accuracy of upper body and lower body posture prediction for each of the 13 participants.

The KAN model shows upper body accuracy between 88.89% and 99.79%, with a mean of $97.03\%(\pm 2.98)$. And lower body accuracy from 73.33% to 100%, averaging $92.11\%(\pm 8.12\%)$. Both models exhibit variability in performance across different participants, with some individuals yielding perfect accuracy for certain posture predictions.

Table 5 presents the time measurements for various stages of the posture recognition process, including data preprocessing and model inference times for both KAN and MLP models.

TABLE 5: Time measurements for various stages of the posture recognition process.

Process Stage	Average Time
MediaPipe processing per frame	93.56 ms
CSV generation per frame	0.89 ms
KAN inference per user and frame	$15.38\mu s$ ($\pm 2.29\mu s$)
MLP inference per user and frame	$2.21\mu s$ ($\pm 0.31\mu s$)

V. DISCUSSION

Results have shown that KANs provides better accuracy results with lower parameters than MLPs for the dataset and problem proposed in this article. Given the type of model evaluation performed in which different participants are tested with models trained using data from other participants, a conclusion that can be obtained is that it can generalize better, which is congruent with original authors claims. Despite KAN showing better generalization abilities for this problem, MLP also shows really good results, which probably could be improved modifying its architecture by testing different activation functions or using dropout layers, but the purpose of this work was comparing vanilla MLPs with KANs. It also should be noted that MLP model is faster for inference, despite having more parameters, which may change in the future if KAN implementations become more efficient. While KANs show promise, their real-world utility and interpretability, especially in larger models, remain areas for further investigation. The parameterization of KANs, particularly in terms of grid size and spline order, offers unique optimization opportunities but also introduces new complexities in model design.

Another relevant contribution is the dataset itself, which is being shared to allow other researchers to work with it. The main limitation of the dataset that it only contains MediaPipe processed CSVs. Its reduces possibilities of other researchers, and because, after some testing and visualizations of MediaPipe outputs, we have verified that it has considerable issues estimating depth. However, as the results prove, that limitation can be overcomed in model training. The main strength of this dataset is the inclusion of a manual labelling of individual frames, that have undergone a peer-review process to ensure curated and correct labels. This meticulous curation process significantly enhances the dataset's reliability and value for posture recognition research. The two label approach also contributes about how posture recognition should be analyzed taking into account different segments of the body according to ISO 11226-2000. In addition, our solution based on a multi-task learning approach fits perfectly the needs of the problem and is more efficient than training more models.

Regarding the state of the art in automatic posture recognition, Table 6 highlights several key improvements of our work. Our system classifies a more comprehensive range of postures (35 distinct combinations) and considers upper and lower body segments separately, offering a more nuanced analysis than most existing studies. While some studies report higher accuracies, it's crucial to note the differences in evaluation methodologies. Our use of LOSO cross-validation provides a more rigorous assessment of model generalization compared to the simpler train/test splits or k-fold validations used in other studies, particularly important when dealing with inter-subject variability.

Our approach balances accessibility and accuracy by combining a standard camera with MediaPipe for pose estimation, making it more viable for widespread adoption compared to systems using specialized sensors. The use of KANs represents a novel approach in this field, achieving competitive accuracy with significantly fewer parameters (4,320) compared to CNN-based models with hundreds of thousands or millions of parameters. This efficiency in parameter use could lead to more lightweight and deployable models.

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TABLE 6: Comparison of sitting posture recognition studies using various approaches.

Study	Sensor Type	Number of Postures	Accuracy	Model type	Validation method	Size
Chen [17]	Camera + OpenPose	2	> 90%	CNN	Training accuracy	17 layers
Estrada et al. [18]	Smartphone Camera + MediaPipe	2	85.18% - 92.07%	Decision Tree	70/30 split + cross-validation	Missing
Jiao et al. [19]	Camera	3	99.57%	LMSPNet (CNN)	Train/test split	220,000 parameters
Gupta et al. [20]	15 PSC sensors	8	99.30%	CNN	60/20/20 split	Missing
Lin et al. [22]	LiDAR	42	99.71%	Modified EfficientNet Lite2	10-fold cross-validation	< 6.1M parameters
Xu and Liu [23]	FMCW radar	4	98.75%	CNN	Missing	Missing
Nishitha et al. [25]	Camera	6	CNN: 92.5%, RCNN: 95%	CNN, RCNN	Missing	CNN: 2 conv layers, RCNN: 41M
Our study	Camera + MediaPipe	35 (5 upper and 7 lower)	94.57%	KAN	LOSO cross-validation	4,320 parameters

However, it's important to note that direct comparisons of model sizes are challenging due to inconsistent reporting across studies and the unknown parameter count of the MediaPipe model. Despite this, our work contributes significantly to the field through its comprehensive classification, rigorous evaluation methodology, and efficient use of a novel neural network architecture, paving the way for more effective and practical posture recognition systems.

Our study makes several key contributions to the field of automatic posture recognition. Firstly, the application of KANs to a practical case of posture recognition demonstrates their potential in real-world scenarios. Secondly, our system achieves excellent results using a rigorous LOSO evaluation method, which provides a more realistic assessment of the model's generalization capabilities compared to traditional cross-validation techniques. Finally, our dual upper-lower labeling system offers a more comprehensive approach to posture analysis, aligning with established ergonomic standards and potentially improving the practical applicability of our results in workplace and healthcare settings.

VI. CONCLUSION

This study provides a comprehensive comparison between Kolmogorov-Arnold Networks (KANs) and classic Multilayer Perceptrons (MLPs) for sitting posture recognition. The results demonstrate that KANs can achieve superior generalization with fewer parameters, while both models exhibit fast inference times suitable for real-time applications. Both KANs and MLPs achieved high accuracy, exceeding 91% in LOSO cross-validation, indicating their potential for robust performance across diverse populations.

A significant contribution of this research is the implementation of a multi-task learning approach for simultaneous upper and lower body posture recognition. This method enhances the efficiency of posture recognition systems and provides a more comprehensive analysis of sitting postures, opening up new possibilities for detailed ergonomic assessments in various settings.

The high accuracy achieved by both models has significant implications for improving ergonomics in workplace and home environments. By providing accurate, real-time feedback on posture, these systems can play a crucial role in preventing musculoskeletal disorders and promoting healthier sitting habits, potentially impacting public health in our increasingly sedentary society.

Future work should focus on expanding the dataset to include a wider range of body types and sitting environments, and conducting long-term studies to assess the impact on user behavior and health outcomes. Additionally, exploring privacy-preserving techniques for posture recognition could address potential concerns about data security and personal privacy in real-world implementations of these systems.

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DATA AVAILABILITY

THE DATA WILL BE AVAILABLE IN THE FINAL VERSION OF THIS PAPER.

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